ADITYA RASAL

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AI Engineer | Computer Vision & Machine Learning | AI-Driven Project Management

PROFESSIONAL SUMMARY

AI Engineer specializing in Computer Vision, with hands-on experience designing, training, and deploying production ML pipelines end-to-end from model development to full-stack deployment tooling and internal vision platforms. Combines this technical delivery with project management experience: leading cross-functional teams of 150+, owning budgets and timelines, and applying Agile/Scrum and stakeholder-management practices across concurrent initiatives. Bachelor of Computer Science Engineering candidate, CGPA 9.1/10, graduating 2027.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, SQL
AI Engineering & Machine Learning: PyTorch (CUDA), Model Training & Evaluation, scikit-learn, Ultralytics YOLO, NumPy, Pandas
Computer Vision: OpenCV, Object Detection, Image Segmentation (SAM-assisted), Real-Time Video Analytics, YOLOv8/YOLO11/YOLO26
Generative AI & LLMs: Prompt Engineering, LLM APIs (Claude, OpenAI), LangChain, LangGraph, Hugging Face, NLP Fundamentals
Backend Development: FastAPI, Django, Flask, Node.js, REST APIs, PostgreSQL, SQL, SQLite
Frontend Development: React, Vue.js, JavaScript, HTML5, CSS3, Responsive Design
Project Management & Tools: Agile/Scrum, Jira, Git/GitHub, Stakeholder Management, Risk Mitigation, Cross-Functional Team Leadership

PROFESSIONAL EXPERIENCE

Computer Vision Project Intern, Mahindra & Mahindra **May 2026 – Aug 2026**
Nashik Plant 1 *Nashik, Maharashtra, India*

- Delivered a production-deployed YOLO-based crack-detection pipeline in under three months, integrating computer vision models with live PLC infrastructure on a manufacturing line, achieving 85% detection accuracy at 25 FPS.
- Automated real-time shop-floor alerting by linking defect detection to hardware hooter activation and a live monitoring web application, cutting inspection response time by 70%.
- Built "Master Launcher," a centralized dashboard managing 30 concurrent video analytics pipelines, optimizing GPU and system resource utilization plant-wide.
- Architected an internal AutoML platform enabling non-technical operators to extract frames, annotate datasets, and train custom YOLO models independently.

PROJECTS

Vision Studio | Offline Computer Vision Model Studio
Tech Stack: Python, FastAPI, PyTorch, Ultralytics YOLO, OpenCV, PyWebView, JavaScript

- Engineered a fully offline desktop platform for end-to-end CV model lifecycle: dataset creation, annotation, and GPU-accelerated training with zero cloud dependency.
- Built a multi-task annotation engine (Detection, SAM-assisted Segmentation, Classification) with automated frame extraction from RTSP/USB feeds and one-click export, TorchScript, and OpenVINO.

Vision Flow | Edge AI Live Detection & Monitoring Platform
Tech Stack: Python, FastAPI, PyTorch, YOLOv8/YOLO11/YOLO26, OpenCV, SQLite (WAL), SSE, ReportLab

- Architected a production edge AI inference system handling concurrent multi-camera streams (USB, RTSP, QR-paired mobile) for real-time plant and perimeter monitoring at ~45ms average latency.
- Designed an active-learning feedback loop that routes false detections back into training data and auto-generates tamper-evident PDF compliance reports, improving accuracy by ~20% over time.

LEADERSHIP & PROJECT MANAGEMENT EXPERIENCE

Pimpri-Chinchwad University, Pune, Maharashtra, India

Cultural President, Student Council **Aug 2026 – Present**

- Direct a 150+ member student volunteer workforce across major campus initiatives (IDEATHON 2.0, Pitch Fest), consistently delivering under budget.

Event Lead, Anantam Cultural Festival **Jan 2024 – Jan 2026**

- Led a 2-day university festival across 3 consecutive editions, managing 200+ team members and coordinating budget, timelines, and vendor/stakeholder relationships for 6,000+ attendees with 100% on-time delivery.

EDUCATION

Bachelor of Computer Science Engineering & Technology **Aug 2023 – May 2027(Expected)**
Pimpri-Chinchwad University, Pune, Maharashtra, India *CGPA: 9.1 / 10*

Higher Secondary Certificate (HSC), Science **Aug 2021 – May 2023**
GD Sawant College of Science, Nashik, Maharashtra, India *68%*

CERTIFICATIONS

Google Project Management Certificate (Google) | IBM Machine Learning Certificate (IBM) | Agile with Atlassian Jira (Atlassian)